HCHS/SOL - TB/DM

Adriana Gutierrez Yllu

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Reading Files: HCHS/SOL V1

```
#Be sure to replace the file path with the appropriate file path on your own computer.

# >>>>> VISIT 1 <<<<<<

library(readr)

base_dir <- "/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 Emory/Rollins/thesis/Tesis - Rollins/

file_paths <- c(
  file.path(base_dir, "part_derv_lad1.csv"),
  file.path(base_dir, "piea_lad1.csv"),
  file.path(base_dir, "rsea_lad1.csv"),
  file.path(base_dir, "mhea_lad1.csv"),
  file.path(base_dir, "muea_lad1.csv"),
  file.path(base_dir, "laba_lad1.csv"),
  file.path(base_dir, "anta_lad1.csv"),
  file.path(base_dir, "sbpa_lad1.csv")
)

for (i in seq_along(file_paths)) {
  oname <- paste0("df", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: HCHS/SOL V1

```
#merging the data set
data_sol_v1 <- merge(df1, df2,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df3,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df4,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df5,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df6,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df7,  by ="PID", all = T)
data_sol_v1 <- merge(data_sol_v1, df8,  by ="PID", all = T)

# cleaning
```

```r
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, file_paths, i, oname)

attach(data_sol_v1)

#creating new data frame which only has the variables of interest
df_sol_v1 <- data.frame(
  id_study = 'HCHS_SOL_V1',
  id       = PID,

  psu1     = as.numeric(PSU_ID),       # check when merging
  psu2     = as.numeric(PSU_CNT),

  stratum1 = as.numeric(STRAT),
  stratum2 = as.numeric(STRAT_CNT),

  psu_weight1 = as.numeric(WEIGHT_FINAL_NORM_OVERALL),
  psu_weight2 = as.numeric(WEIGHT_FINAL_EXPANDED),
  psu_weight3 = as.numeric(WEIGHT_FINAL_NORM_CENTER),

  # demographics
  sex     = PIEA1,
  age     = as.numeric(AGE),
  race    = PIEA6,
  edu_adult = PIEA19A,
  edu_years  = as.numeric(PIEA17), #years
  edu_child = NA,
  ethnicity = NA,

  #diabetes
  self_db      = MHEA16, #diabetes ever
  age_told_db  = as.numeric(MHEA16A),
  drug_insu_db = MHEA16C,
  drug_pill_db = MUEA33C, #but only last 4 weeks
  family_hx_db = NA,

  #labs
  fasting_minutes = NA,
  fasting_hours   = NA,
  fasting_8h      = NA,
  fasting_12h     = NA,
  fglu         = as.numeric(LABA70), units_fglu  = 'mg/dL',
  hba1c        = as.numeric(LABA72), units_hba1c = 'NA',
  pogtt          = NA, units_pogtt   = 'NA',
  insulin      = NA, units_insulin = 'NA',

  creatinine_urine = as.numeric(LABA76), creatinine_urine_units = 'mg/dL',

  total_cholesterol        = as.numeric(LABA66), units_cholesterol = NA,
  ldl                      = as.numeric(LABA69), units_ldl = 'mg/dL',

  #tb
  tb_history = RSEA50,
```

```
  age_told_tb  = NA,
  tb_treatment = RSEA51,
  tb_skin_test = RSEA52, #self-reported
  tb_skin_test_result = RSEA53, #self-reported
  tb_medication_use = RSEA54,

  # Measurements
  weight                  = as.numeric(ANTA4), units_weight = 'kg',
  height                  = as.numeric(HEIGHT), units_height = 'cm',
  bmi                     = as.numeric(BMI), units_bmi = 'KG/M**2',
  waist                   = as.numeric(ANTA10A), units_waist = 'cm',          # just 1 measure
  hip                     = as.numeric(ANTA10B), units_waist = 'cm',
  # These will be converted into blood_pressure_sys and blood_pressure_diast later
  blood_pressure_sys2      = as.numeric(SBPA11), units_sbp = 'mmHg',
  blood_pressure_sys3      = as.numeric(SBPA14), units_sbp = 'mmHg',
  blood_pressure_diast2    = as.numeric(SBPA12),
  blood_pressure_diast3    = as.numeric(SBPA15)
)

detach(data_sol_v1)
```

Reading Files: HCHS/SOL V2

```
# >>>>> VISIT 2 <<<<<<
# without demographic variables -> meaning this is a follow up after ~7 years

library(readr)

#Be sure to replace the file path with the appropriate file path on your own computer.

base_dir <- "/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 Emory/Rollins/thesis/Tesis - Rollins/

file_paths <- c(
  file.path(base_dir, "mhe_v2_lad1.csv"),
  file.path(base_dir, "mue_v2_lad1.csv"),
  file.path(base_dir, "hce_v2_lad1.csv"),
  file.path(base_dir, "lab_derv_v2_lad1.csv"),
  file.path(base_dir, "ant_v2_lad1.csv"),
  file.path(base_dir, "part_derv_v2_lad1.csv"),
  file.path(base_dir, "sbp_v2_lad1.csv")
)

for (i in seq_along(file_paths)) {
  oname <- paste0("dfv2_", i)
  assign(oname, read_csv(file_paths[i]), envir = .GlobalEnv)
}
```

Cleaning: HCHS/SOL V2

```r
#merging the data set
data_sol_v2 <- merge(dfv2_1, dfv2_2,  by ="PID", all = T)
data_sol_v2 <- merge(data_sol_v2, dfv2_3,  by ="PID", all = T)
data_sol_v2 <- merge(data_sol_v2, dfv2_4,  by ="PID", all = T)
data_sol_v2 <- merge(data_sol_v2, dfv2_5,  by ="PID", all = T)
data_sol_v2 <- merge(data_sol_v2, dfv2_6,  by ="PID", all = T)
data_sol_v2 <- merge(data_sol_v2, dfv2_7,  by ="PID", all = T)

# cleaning
rm(dfv2_1, dfv2_2, dfv2_3, dfv2_4, dfv2_5, dfv2_6, dfv2_7, file_paths, i, oname)

attach(data_sol_v2)

#creating new data frame which only has the variables of interest
df_sol_v2 <- data.frame(
  id_study = rep("HCHS_SOL_V2", length(PID)),
  id       = PID,

  psu1     = NA,       # check when merging
  psu2     = NA,

  stratum1 = NA,
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  sex     = NA,
  age     = NA,
  race    = NA,
  edu_adult = NA, #level of education
  edu_years = NA,
  edu_child = NA,
  ethnicity = NA,

  #diabetes
  self_db      = NA, #diabetes ever
  age_told_db  = NA,
  drug_insu_db = NA,
  drug_pill_db = NA, #but only last 4 weeks
  family_hx_db = NA,

  #labs
  fasting_minutes = NA,
  fasting_hours   = NA,
  fasting_8h      = NA,
  fasting_12h     = NA,
  fglu         = as.numeric(LABA70), units_fglu  = 'mg/dL',
  hba1c        = as.numeric(LABA72), units_hba1c = '%',
```

```
  pogtt        = as.numeric(LABA71), units_pogtt  = 'NA',
  insulin      = as.numeric(LABA96), units_insulin = 'uU/mL',

  creatinine_urine = as.numeric(LABA76), creatinine_urine_units = 'mg/dL',

  total_cholesterol        = as.numeric(LABA66), units_cholesterol = 'mg/dL',
  ldl                      = as.numeric(LABA69), units_ldl = 'mg/dL',

  #tb - asked "since V1.."
  tb_history = MHE35,
  age_told_tb  = NA,
  tb_treatment = MHE35A,
  tb_skin_test = MHE36, #self-reported
  tb_skin_test_result = MHE36A, #self-reported
  tb_medication_use = MHE36B,

  # Measurements
  weight                   = as.numeric(ANT4), units_weight = 'kg',
  height                   = as.numeric(ANT2), units_height = 'cm',
  bmi                      = as.numeric(BMI_V2), units_bmi = 'KG/M**2',
  waist                    = as.numeric(ANT10A), units_waist = 'cm',    # just 1 measure
  hip                      = as.numeric(ANT10B), units_waist = 'cm',
  # These will be converted into blood_pressure_sys and blood_pressure_diast later
  blood_pressure_sys2       = as.numeric(SBP11), units_sbp = 'mmHg',
  blood_pressure_sys3       = as.numeric(SBP14), units_sbp = 'mmHg',
  blood_pressure_diast2     = as.numeric(SBP12),
  blood_pressure_diast3     = as.numeric(SBP15)
)

detach(data_sol_v2)
```

Re-coding V1 and V2 datasets

```
recode_sol <- function(df) {
  df %>%
    mutate(

      # sex
      sex = case_when(
        sex == 1 ~ 1,  # men
        sex == 2 ~ 2,  # women
        TRUE ~ NA_real_
      ),

      # yes/no variables
      across(
        c(self_db, drug_insu_db, drug_pill_db,
          tb_history, tb_treatment, tb_skin_test),
        ~ case_when(
          . == 1 ~ 1,
          . == 0 ~ 0,
```

```
      . %in% c(2, 9) ~ 9,
      TRUE ~ NA_real_
    )
  ),

  # TB skin test result
  tb_skin_test_result = case_when(
    tb_skin_test_result == 2 ~ 0,                  # negative
    tb_skin_test_result == 1 ~ 1,                  # positive
    tb_skin_test_result %in% c("Q", 9) ~ 9,        # DK
    tb_skin_test_result == "S" ~ NA_real_,
    TRUE ~ NA_real_
  ),

  # TB medication use after positive test
  tb_medication_use = case_when(
    tb_medication_use == 0 ~ 0,
    tb_medication_use == 1 ~ 1,
    tb_medication_use == 9 ~ 9,
    tb_medication_use == "S" ~ NA_real_,
    TRUE ~ NA_real_
  ),

  # race (among hispanics)
  race = case_when(
    race %in% c(1, 2, 3, 4, 5, 6) ~ 2,  # hispanic
    race %in% c("Q", 9) ~ 9,            # DK
    TRUE ~ NA_real_                     # missing
  ),

  # ethnicity (all Hispanics)
  ethnicity = case_when(
    TRUE ~ 2                 # Hispanics
  ),

  # education
  edu_adult = case_when(
    edu_adult %in% c(1, 2) ~ 1,    # < HS
    edu_adult == 3 ~ 2,            # HS
    edu_adult %in% c(4, 5) ~ 3,    # some college+
    edu_adult %in% c(6, "Q") ~ 9,  # refused / DK
    edu_adult == "S" ~ 9,
    TRUE ~ 9
  )
  )
}

# Recode datasets (SOL V1, V2)
df_sol_v1 <- recode_sol(df_sol_v1)
df_sol_v2 <- recode_sol(df_sol_v2)
```

Diabetes definitions

```r
# Diabetes: Self reported / labs (fasting glucose, HbA1c, oral glucose tolerance test)
create_self_lab_dm <- function(dm) {
  dm %>%
    mutate(
      self_lab_dm = case_when(
        self_db == 1 | fglu >= 126 | hba1c >= 6.5 | pogtt >= 200 ~ 1,
        !is.na(self_db) | !is.na(fglu) | !is.na(hba1c) | !is.na(pogtt) ~ 0,
        is.na(self_db) & is.na(fglu) & is.na(hba1c) & is.na(pogtt) ~ NA_real_
      )
    )
}

df_sol_v1 <- create_self_lab_dm(df_sol_v1)
df_sol_v2 <- create_self_lab_dm(df_sol_v2)
```

Generating same SBP and DBP as other datasets

```r
# Average of blood pressure measurement (systolic and diastolic #2 + #3 / 2)
process_bp <- function(df) {
  df$blood_pressure_sys   <- (df$blood_pressure_sys2 + df$blood_pressure_sys3) / 2
  df$blood_pressure_diast <- (df$blood_pressure_diast2 + df$blood_pressure_diast3) / 2

  df <- subset(df, select = -c(blood_pressure_sys2,
                               blood_pressure_sys3,
                               blood_pressure_diast2,
                               blood_pressure_diast3))
  return(df)
}

df_sol_v1 <- process_bp(df_sol_v1)
df_sol_v2 <- process_bp(df_sol_v2)

# Summary of numeric variables - 1999
summary(df_sol_v1$blood_pressure_sys)
summary(df_sol_v1$blood_pressure_diast)

library(dplyr)

df_sol_v1 <- df_sol_v1 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
```

```
  )


# Summary of numeric variables - df_sol_v2

summary(df_sol_v2$blood_pressure_sys)
summary(df_sol_v2$blood_pressure_diast)


df_sol_v2 <- df_sol_v2 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )
```

Numeric variables

```
# Summary of numeric variables - df_sol_v1
summary(df_sol_v1$weight)
summary(df_sol_v1$height)
summary(df_sol_v1$bmi)
summary(df_sol_v1$waist)
summary(df_sol_v1$hip)
summary(df_sol_v1$ldl)
summary(df_sol_v1$total_cholesterol)


library(dplyr)

df_sol_v1 <- df_sol_v1 %>%
  mutate(
    weight = ifelse(weight >= 12 & weight <= 300, weight, NA),              # Plausible range: 12-300 k
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 67.66 & total_cholesterol <= 773.31,             # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )


# Summary of numeric variables - df_sol_v2
```

```r
summary(df_sol_v2$weight)
summary(df_sol_v2$height)
summary(df_sol_v2$bmi)
summary(df_sol_v2$waist)
summary(df_sol_v2$hip)
summary(df_sol_v2$ldl)
summary(df_sol_v2$total_cholesterol)

df_sol_v2 <- df_sol_v2 %>%
  mutate(
    weight = ifelse(weight >= 12 & weight <= 300, weight, NA),              # Plausible range: 12-300 k
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 67.66 & total_cholesterol <= 773.31,                  # Plausible range: 1.7
      total_cholesterol, NA
    )
  )
```

Labeling both datasets

```r
label_sol <- function(label) {
  label %>%
    mutate(
      # Sex
      sex = factor(sex,
                   levels = c(1, 2),
                   labels = c("Men", "Women")),

      # TB variables
      tb_history = factor(tb_history,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
      tb_treatment = factor(tb_treatment,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      tb_skin_test = factor(tb_skin_test,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      tb_skin_test_result = factor(tb_skin_test_result,
                                   levels = c(1, 0, 9),
                                   labels = c("Positive", "Negative", "Refused/DK")),
      tb_medication_use = factor(tb_medication_use,
                                 levels = c(1, 0, 9),
                                 labels = c("Yes", "No", "Refused/DK")),

      # Diabetes self-report + labs
      self_db = factor(self_db,
```

```
                        levels = c(1, 0, 9),
                        labels = c("Yes", "No", "Refused/DK")),
      drug_insu_db = factor(drug_insu_db,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      drug_pill_db = factor(drug_pill_db,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      self_lab_dm = factor(self_lab_dm,
                           levels = c(1, 0),
                           labels = c("Diabetes", "No Diabetes")),

      # Family history
      family_hx_db = factor(family_hx_db,
                            levels = c(1, 0),
                            labels = c("Yes", "No")),

      # Race/Ethnicity
      race = factor(race,
                    levels = c(0, 1, 2, 3, 9),
                    labels = c("Non-Hispanic Black",
                               "Non-Hispanic White",
                               "Mexican/Other Hispanic",
                               "Other/Multiracial",
                               "Missing")),
      ethnicity = factor(ethnicity,
                         levels = c(1, 2, 9),
                         labels = c("Non-Hispanic", "Hispanic", "Missing")),

      # Education
      edu_adult = factor(edu_adult,
                         levels = c(1, 2, 3, 4, 9),
                         labels = c("< High School",
                                    "High School Grad",
                                    "Some College/AA",
                                    "College+",
                                    "Refused/DK"))
    )
}

df_sol_v1 <- label_sol(df_sol_v1)
df_sol_v2 <- label_sol(df_sol_v2)
```

Exploratory data analysis

```
###==== V1 - Baseline visit =======
#TB (n=423)
table(df_sol_v1$tb_history)

#DM
table(df_sol_v1$self_db)        # (n=1885)
```

```r
table(df_sol_v1$self_lab_dm)    # (n=2386)

#DM and TB
table(df_sol_v1$tb_history, df_sol_v1$self_db)      # (n=72)
table(df_sol_v1$tb_history, df_sol_v1$self_lab_dm)  # (n=89)

###==== Visit V2 =======
#TB (n=65)
table(df_sol_v2$tb_history)

#DM
table(df_sol_v2$self_db) # (n=0) no self report data, didn't ask the question
table(df_sol_v2$tb_history, df_sol_v2$self_lab_dm)  # (n=17) new db diagnosis since last visit


#DM and TB
table(df_sol_v2$tb_history, df_sol_v2$self_db)      # (n=0)
table(df_sol_v2$tb_history, df_sol_v2$self_lab_dm)  # (n=17)
```

New DM and TB variable

```r
# For CARDIA Y00
df_sol_v1 <- df_sol_v1 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,  # both DM and TB
      TRUE ~ 0                                               # everything else
    )
  )

# For CARDIA Y10
df_sol_v2 <- df_sol_v2 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,
      TRUE ~ 0
    )
  )

# Labeling both dataset
label_tb_dm <- function(tb_dm) {
  tb_dm %>%
    mutate(
      # diabetes & tuberculosis
      tb_dm = factor(tb_dm,
                     levels = c(0, 1),
                     labels = c("No", "Yes"))
    )
}

df_sol_v1 <- label_tb_dm(df_sol_v1)
```

```r
df_sol_v2 <- label_tb_dm(df_sol_v2)


#Checking
table(df_sol_v1$tb_dm)
table(df_sol_v2$tb_dm)
```

Preliminary results:

SOL V1

```r
library(gtsummary)

# Select the variables you want to include
vars_to_summarize <- c(
  "age", "sex", "race", "edu_adult", "self_db", "age_told_db", "drug_insu_db", "drug_pill_db",
  "self_lab_dm", "family_hx_db",
  "tb_history", "age_told_tb", "tb_treatment", "fasting_minutes", "fglu", "hba1c", "total_cholesterol",

# Create table1
SOL_V1_table <- df_sol_v1 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_sol_v1),
    missing = "no"
  )

SOL_V1_table <- SOL_V1_table %>%
```

```
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (SOL V1)**")

SOL_V1_table
```

Preliminary results:

SOL V2

```
library(gtsummary)

SOL_V2_table <- df_sol_v2 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_sol_v2),
    missing = "no"
  )

SOL_V2_table <- SOL_V2_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (SOL V2)**")

SOL_V2_table
```

Variables	Summary Statistics (SOL V1)	
	No N = 12,558[1]	**Yes** N = 89[1]
Age	46 (14)	55 (10)
Sex		
Men	5,243 / 12,557 (42%)	35 / 89 (39%)
Women	7,314 / 12,557 (58%)	54 / 89 (61%)
Race		
Non-Hispanic Black	0 / 12,372 (0%)	0 / 89 (0%)
Non-Hispanic White	0 / 12,372 (0%)	0 / 89 (0%)
Mexican/Other Hispanic	7,841 / 12,372 (63%)	53 / 89 (60%)
Other/Multiracial	0 / 12,372 (0%)	0 / 89 (0%)
Missing	4,531 / 12,372 (37%)	36 / 89 (40%)
Education Level		
< High School	2,821 / 12,558 (22%)	28 / 89 (31%)
High School Grad	4,771 / 12,558 (38%)	42 / 89 (47%)
Some College/AA	4,571 / 12,558 (36%)	18 / 89 (20%)
College+	0 / 12,558 (0%)	0 / 89 (0%)
Refused/DK	395 / 12,558 (3.1%)	1 / 89 (1.1%)
Self-reported diabetes		
Yes	1,813 / 12,499 (15%)	72 / 89 (81%)
No	10,686 / 12,499 (85%)	17 / 89 (19%)
Refused/DK	0 / 12,499 (0%)	0 / 89 (0%)
Age told DM	45 (13)	44 (12)
Diabetes Mellitus on insulin		
Yes	396 / 1,806 (22%)	22 / 72 (31%)
No	1,410 / 1,806 (78%)	50 / 72 (69%)
Refused/DK	0 / 1,806 (0%)	0 / 72 (0%)
Diabetes Mellitus on Oral med		
Yes	1,395 / 8,252 (17%)	50 / 78 (64%)
No	6,851 / 8,252 (83%)	27 / 78 (35%)
Refused/DK	6 / 8,252 (<0.1%)	1 / 78 (1.3%)
DM self-reported and by lab def		
Diabetes	2,297 / 12,556 (18%)	89 / 89 (100%)
No Diabetes	10,259 / 12,556 (82%)	0 / 89 (0%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
Yes	334 / 12,352 (2.7%)	89 / 89 (100%)
No	11,974 / 12,352 (97%)	0 / 89 (0%)
Refused/DK	44 / 12,352 (0.4%)	0 / 89 (0%)
Age told TB	NA (NA)	NA (NA)
TB treatment ever		
Yes	251 / 380 (66%)	66 / 89 (74%)
No	112 / 380 (29%)	23 / 89 (26%)
Refused/DK	17 / 380 (4.5%)	0 / 89 (0%)
Fasting (minutes)	NA (NA)	NA (NA)
Fasting Glucose (mg/dl)	105 (37)	149 (65)
HbA1c (%)	5.89 (1.30)	7.65 (2.00)
Total cholesterol (mg/dL)	199 (44)	198 (53)
LDL (mg/dL)	123 (37)	120 (44)
Weight (lbs)	78 (17)	82 (18)
Height (cm)	162 (9)	160 (9)

Variables	Summary Statistics (SOL V2)	
	No N = 8,821[1]	**Yes** N = 17[1]
Age	NA (NA)	NA (NA)
Sex		
Men	0 / 0 (NA%)	0 / 0 (NA%)
Women	0 / 0 (NA%)	0 / 0 (NA%)
Race		
Non-Hispanic Black	0 / 0 (NA%)	0 / 0 (NA%)
Non-Hispanic White	0 / 0 (NA%)	0 / 0 (NA%)
Mexican/Other Hispanic	0 / 0 (NA%)	0 / 0 (NA%)
Other/Multiracial	0 / 0 (NA%)	0 / 0 (NA%)
Missing	0 / 0 (NA%)	0 / 0 (NA%)
Education Level		
< High School	0 / 8,821 (0%)	0 / 17 (0%)
High School Grad	0 / 8,821 (0%)	0 / 17 (0%)
Some College/AA	0 / 8,821 (0%)	0 / 17 (0%)
College+	0 / 8,821 (0%)	0 / 17 (0%)
Refused/DK	8,821 / 8,821 (100%)	17 / 17 (100%)
Self-reported diabetes		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Age told DM	NA (NA)	NA (NA)
Diabetes Mellitus on insulin		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
Diabetes Mellitus on Oral med		
Yes	0 / 0 (NA%)	0 / 0 (NA%)
No	0 / 0 (NA%)	0 / 0 (NA%)
Refused/DK	0 / 0 (NA%)	0 / 0 (NA%)
DM self-reported and by lab def		
Diabetes	2,058 / 8,752 (24%)	17 / 17 (100%)
No Diabetes	6,694 / 8,752 (76%)	0 / 17 (0%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
Yes	48 / 8,586 (0.6%)	17 / 17 (100%)
No	8,538 / 8,586 (99%)	0 / 17 (0%)
Refused/DK	0 / 8,586 (0%)	0 / 17 (0%)
Age told TB	NA (NA)	NA (NA)
TB treatment ever		
Yes	22 / 46 (48%)	11 / 14 (79%)
No	24 / 46 (52%)	3 / 14 (21%)
Refused/DK	0 / 46 (0%)	0 / 14 (0%)
Fasting (minutes)	NA (NA)	NA (NA)
Fasting Glucose (mg/dl)	111 (41)	175 (79)
HbA1c (%)	6.13 (1.35)	8.76 (2.40)
Total cholesterol (mg/dL)	193 (41)	192 (48)
LDL (mg/dL)	116 (35)	111 (43)
Weight (lbs)	79 (17)	86 (21)
Height (cm)	162 (9)	163 (10)